10. Analyzing and Interpreting Financial Statements (Ch 5)

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Part 1: Vertical and Horizontal Analysis

10.1 Financial Statement Analysis

Financial Statement Analysis

- Identifies relationships and **trends within financial statement** numbers **over time** and **compared to other companies** or industry averages.

- Helps users (investors, creditors, managers) interpret and understand company performance, financial health, and risks.

- Enables meaningful comparisons to assess investment risk, credit risk, and growth potential.

Assessing the Business Environment First

- Before interpreting financial ratios, we must understand the company's context (industry, operations, risks). This helps ensure ratios are interpreted correctly.
- Understand the company's stage and strategy.
  - Identify where the company is in its life cycle (start-up, growth, mature)
- Analyze what the company sells and to whom.
  - Evaluate products, customer demographics, and demand trends
- Evaluate competitive position.
  - Determine how the firm compares to rivals and whether new entrants threaten it
- Assess internal resources & capabilities.
  - Review suppliers, labor, technology, and operational support systems
- Review financing and capital structure.
  - Understand reliance on debt, ability to raise capital, and financial risk
- Consider regulatory and political environment.
  - Identify laws and government influence that may affect operations

10.2 Vertical Analysis: Common-size financial statements

- A financial statement analysis method that expresses each item in a financial statement **as a percentage of a designated base amount**
  - Income statement items → % of **net sales**
  - Balance sheet items → % of **total assets**
- Also called **common-size financial statements**, which facilitate meaningful comparisons across **companies of varying sizes** and within **different time periods**.
- By restating financial data in proportional terms, vertical analysis reveals important insights such as the composition of assets, liabilities, and equity on the balance sheet, as well as cost structure and profit margins on the income statement.

Vertical Analysis of Balance Sheet (Exhibit 5.1)

EXHIBIT 5.1 PepsiCo Comparative Balance Sheets

PEPSICO, INC.
Balance Sheets and Common-Size Balance Sheets
December 30, 2020 and December 31, 2019

	As Reported ($ millions)		As a Percentage of Total Assets	
	2020	**2019**	**2020**	**2019**
Assets				
Current assets				
Cash and cash equivalents	$ 8,185	$ 5,509	8.8%	7.0%
Short-term investments	1,366	229	1.5%	0.3%
Accounts and notes receivable, net	8,404	7,822	9.0%	10.0%
Inventories	4,172	3,338	4.5%	4.2%
Prepaid expenses and other current assets	874	747	0.9%	1.0%
Total current assets	23,001	17,645	24.7%	22.5%
Property, plant, and equipment, net	21,369	19,305	23.0%	24.6%
Amortizable intangible assets, net	1,703	1,433	1.8%	1.8%
Goodwill	18,757	15,501	20.2%	19 .7%
Other indefinite-lived intangible assets	17,612	14,610	19.0%	18.6%
Investments in noncontrolled affiliates	2,792	2,683	3.0%	3.4%
Deferred income taxes	4,372	4,359	4.7%	5.5%
Other assets	3,312	3,011	3.6%	3.8%
Total assets	$92,918	$78,547	100.0%	100.0%
Liabilities and equity				
Current liabilities				
Short-term debt obligations	$ 3,780	$ 2,920	4.1%	3.7%
Accounts payable and other current liabilities	19,592	17,541	21.1%	22.3%
Total current liabilities	23,372	20,461	25.2%	26.0%
Long-term debt obligations	40,370	29,148	43.4%	37.1%
Deferred income taxes	4,284	4,091	4.6%	5.2%
Other liabilities	11,340	9,979	12.2%	12.7%
Total liabilities	79,366	63,679	85.4%	81.1%
Total equity	13,552	14,868	14.6%	18.9%
Total liabilities and equity	$92,918	$78,547	100.0%	100.0%

Vertical Analysis of Income Statement (Exhibit 5.2)

PEPSICO, INC.
Income Statements and Common-Size Income Statements
Fiscal Years Ended December 30, 2020 and December 31, 2019

	As Reported ($ millions)		As a Percentage of Net Revenue	
	2020	**2019**	**2020**	**2019**
Net revenue	$70,372	$67,161	100.0%	100.0%
Cost of sales	31,797	30,132	45.2%	44.9%
Gross profit	38,575	37,029	54.8%	55.1%
Selling, general, and administrative expenses	28,495	26,738	40.5%	39.8%
Operating profit	10,080	10,291	14.3%	15.3%
Other pension and retiree medical benefits expense	117	(44)	0.2%	(0.1)%
Net interest expense and other	(1,128)	(935)	(1.6)%	(1.4)%
Income before income taxes	9,069	9,312	12.9%	13.8%
Provision for taxes	1,894	1,959	2.7%	2.9%
Net income	$ 7,175	$ 7,353	10.2%	10.9%

$31.797 / $70.372 = 0.452

10.3 Horizontal Analysis

- Compares financial data across periods (**year-to-year** or **quarter-to-quarter**).
- Shows trends over time in revenue, expenses, income, assets, etc.
- Useful for forecasting future performance based on historical trends.
- Expresses change in both:
  - Dollar amount change (Current year − Prior year)
  - Percentage change $\left(\frac{\text{Change}}{\text{Prior year}}\right) \times 100$
- Key Considerations:
  - Look at both % change and dollar change (a tiny base can create misleading % changes).
  - Large % changes often occur when prior year amounts were very small or near zero.

PEPSICO, INC.
Revenue, Operating Profit, and Net Income
($ millions and percent changes)

	2020	2019	2018
Revenue	$70,372	$67,161	$64,661
	4.8%	3.9%	
Operating profit	$10,080	$10,291	$10,110
	−2.1%	1.8%	
Net income	$ 7,175	$ 7,353	$12,559
	−2.4%	−41.5%	

$$\frac{(\$70{,}372 - \$67{,}161)}{\$67{,}161}$$

Part 2: Return on Investment

10.4 ROE, ROA, and ROFL

Return on Equity (ROE)

$$\text{ROE} = \frac{\text{Net income}}{\text{Avereage stockholder's equity}} \quad \text{or} \quad \text{ROE} = \frac{\text{Net income} - \text{Preferred stock dividends}}{\text{Avereage common stockholder's equity}}$$

- Measures how effectively a company **generates** profit from shareholders' investment.
  - It reflects 1) company profitability and 2) efficiency in using equity capital
- Why ROE matters?
  - Used widely by analysts, investors, and managers
  - Warren Buffett focuses on companies with:
    - High ROE
    - Low or reasonable debt (why? → as high debt can increase ROE, to be discussed)
  - High ROE suggests strong ability to generate returns from equity capital
- Cautions
  - ROE can be boosted artificially by using more debt
  - Excessive debt raises risk, including potential bankruptcy

Use the average balance (beginning and ending) from the balance sheet when comparing it to income statement amounts, instead of the year-end balance.

Return on Assets (ROA)

- ROA measures how efficiently a company uses its **assets** to generate profit.
- Focuses on **operating** and **investing** performance, **regardless of financing method**.
  - Excludes the effect of interest since it relates to financing, not operations.
  - Allows comparison across companies with different leverage levels.
- ROA Formula (de-levered ROA)

$$\text{ROA} = \frac{\text{Earnings}^{1}\text{ without interest expense (EWI)}}{\text{Average total assets}}$$

$$= \frac{\text{Net income} + \text{Interest expense} \times (1 - \text{Tax rate})}{\text{Average total assets}}$$

Income before int. exp
− int. expense
− tax expense
Net income

- Why do we add back after-tax interest expense? because:
  - Net income subtracts interest (financing cost)
  - ROA wants to measure performance before financing decisions
  - Interest is tax-deductible → add back interest net of tax
- ROA without considering the financing effects is often called **de-levered ROA**.

[1] In financial reporting, *net income* is the precise GAAP measure. The term *earnings* is often used interchangeably with net income, but in practice it can also refer to other profit measures—such as operating earnings or adjusted earnings—depending on context.

- Alternatively, ROA (Standard) can be computed as follows:

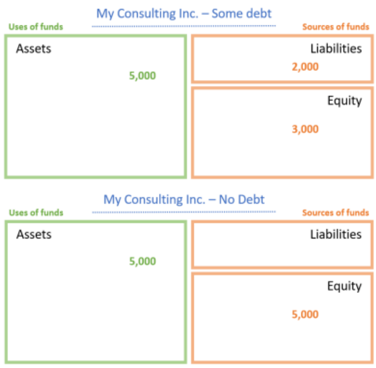
$$\text{ROA} = \frac{\text{Net Income}}{\text{Average total assets}}$$

Term	**ROA (standard)**	**De-levered ROA (covered here)**
Formula	$\frac{\text{Net Income}}{\text{Average total assets}}$	$\frac{\text{Net Income} + \text{after tax effect}}{\text{Average total assets}}$
Includes Debt Impact?	Yes	No
What It Shows	- Profitability after interest cost	- Asset return from operations only before financing effects

Illustration: How de-levering NI removes the effects of capital structure:

	No debt	Some debt
Pretax, pre-interest income	$300	$300
Interest expense	0	(50)
Pretax income	300	250
Taxes (35%)	(105)	(87.5)
Net income	**195**	**162.5**
De-levered Net Income	**$195**	**$195[a]**

[a] after tax effect : NI+ Interest expense ×(1-Tax rate)= [162.5 + 50(1 - .35)] = $195



Return on Financial Leverage (ROFL)

- Measures how much of a company's **return on equity (ROE)** is driven by **financial leverage (use of debt)**.
- **ROFL Formula:** $\text{ROFL} = \text{ROE} - \text{ROA}$
- Shows the **additional return** earned for shareholders from using **debt financing**.
- Captures the portion of ROE that comes **from leverage** rather than operations.

Key Concepts

- **ROE** reflects returns **after financing costs** (includes debt).
- **ROA** reflects returns **before financing costs** (excludes debt).
- The **difference (ROFL)** shows how much financial leverage boosts (or reduces) shareholder returns.

Interpretation

- **Positive ROFL:** Leverage **increases** ROE — debt is being used effectively.
- **Negative ROFL:** Leverage **reduces** ROE — debt costs outweigh its benefits.
- **Higher ROFL = Higher Risk:** More leverage amplifies both gains and losses.

Example (PepsiCo 2020)

- $\text{ROFL} = 50.5\%(\text{ROE}) - 9.5\%(\text{ROA}) = 41.0\%$
- About **41% of PepsiCo's ROE** was due to financial leverage.

10.5 ROA Disaggregation

Disaggregating ROA into Profit Margin × Asset Turnover

- ROA can be broken into two performance drivers:

$$\text{ROA} = \frac{\text{Earnings without interest expense (EWI)}}{\text{Average total assets}} = \underbrace{\frac{\text{EWI}}{\text{Sales revenue}}}_{\substack{\text{Profit Margin} \\ \text{(Profitability)}}} \times \underbrace{\frac{\text{Sales revenue}}{\text{Average total assets}}}_{\substack{\text{Asset Turnover} \\ \text{(Efficiency)}}}$$

ROA = Profit Margin X Asset Turnover

- This helps identify **whether a company's asset performance comes from profitability or efficiency.**

Profit Margin (PM) — Profitability

- Measures **profit earned per dollar of sales** (before financing costs)
- Higher PM = stronger pricing power and cost control
- Influenced by:
  - Gross profit (price vs. cost of goods sold)
  - Operating expenses (wages, marketing, R&D, depreciation)
  - Capacity costs and competitive environment
  - Strategy & discretionary spending (advertising, R&D)

Asset Turnover (AT) — Efficiency

- Measures **sales generated per dollar of assets**
- Higher AT = **more efficient use of assets**
- Driven by:
  - Inventory & receivables management
  - Asset intensity (industry characteristics)
  - Working capital efficiency
- Industry insight:
  - Retail firms: **higher AT** (low fixed asset base)
  - Capital-intensive firms (telecom, utilities): **lower AT** (<1.0)
- PepsiCo 2020 AT = 0.821
  - Each $1 of assets generated $0.82 in sales

Why This Matters

- **ROA increases** when a company:
  - Earns strong profits **(high PM)**, or
  - Efficiently uses assets **(high AT)** — ideally both
- Helps diagnose whether performance changes are due to:
  - Margins falling or costs rising, or
  - Assets not generating enough sales

Trade-Off Between Profit Margin and Asset Turnover

ROA = Profit Margin × Asset Turnover

- A company can achieve the same ROA through **different combinations** of profitability and efficiency.
- Companies face a **trade-off** between earning **higher profit margins** and achieving **higher asset turnover**. → Increasing one often means reducing the other.

Pepsi Co. Example

Year	ROA	Profit Margin	Asset Turnover
2019	10.50%	12.22%	0.860
2020	9.50%	11.53%	0.821

Why the Trade-Off Happens

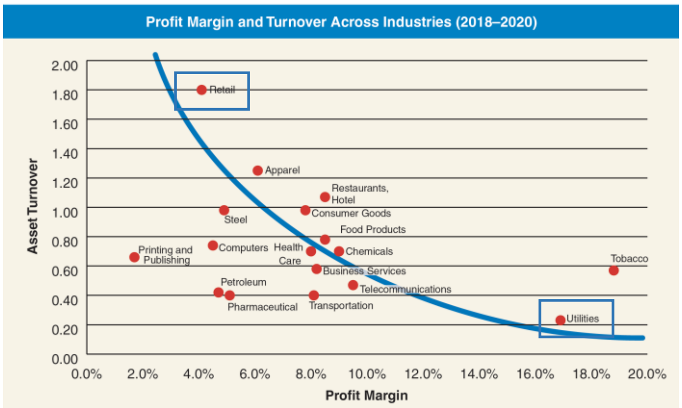
- Profit Margin (PM)
  - Influenced by pricing, costs, and competition.
    → Higher margins often come from **premium pricing or lower costs**, which may require larger investments (lower efficiency).

- Asset Turnover (AT)
  - Measures how efficiently assets generate sales.
    → Higher turnover usually comes from **low margins and high volume**, common in mass-market or retail models.

- Industry Influence
  - High-margin, low-turnover → **capital-intensive** industries (e.g., tech, utilities).
  - Low-margin, high-turnover → **high-volume** industries (e.g., retail, consumer goods).

- Firms can target better ROA by
  - Improving **profit margins** through pricing or cost control, **or**
  - Increasing **asset turnover** through efficiency and faster asset utilization.

- The optimal mix depends on the **business model, strategy, and industry structure**.
  - **Retail Industry:** Intense competition leads to **low profit margins**, but **high asset turnover** due to large sales volume and efficient asset use.
  - **Utility Industry:** Limited competition allows for **high profit margins**, but **low asset turnover** because of **heavy capital investment requirements**.



Further Disaggregation of Profit Margin and Asset Turnover

- Analysts often break down ROA into more detailed ratios to understand **specific drivers of profitability and efficiency, which** helps pinpoint **why** changes occur – whether due to cost management, pricing, or asset use.

Profit Margin (PM) Components

a. Gross Profit Margin (GPM)

$$\text{GPM} = \frac{\text{Sales revenue} - \text{Cost of goods sold}}{\text{Sales revenue}}$$

- Shows how much profit remains after covering the **cost of goods sold (COGS)**.
- Reflects **product pricing**, **production efficiency**, and **cost control**.
- Higher GPM = more revenue retained to cover operating expenses.
- **PepsiCo's GPM (2020): 54.8%** → About 55¢ of each sales dollar is gross profit.

b. Expense-to-Sales Ratio (ETS)

$$\text{ETS} = \frac{\text{SG\&A, R\&D, Advertising, and other expenses}}{\text{Sales revenue}}$$

- Indicates what portion of each sales dollar is spent on **operating expenses**.
- Includes selling, general & administrative (SG&A), marketing, R&D, and advertising.
- Helps track **cost efficiency and spending trends**.
- **PepsiCo's SG&A ETS (2020): 40.5%** → About 41¢ of each sales dollar goes to marketing & administrative costs.

Asset Turnover (AT) Components

- To better understand how efficiently a company uses assets, analysts focus on specific **turnover ratios**:

a. Accounts Receivable Turnover (ART)

$$\text{ART} = \frac{\text{Sales revenue}}{\text{Average accounts receivable}}$$

- Measures how quickly receivables are collected.
- High ART = customers pay faster; low ART = slower collections.
- **PepsiCo's ART (2020): 8.7 times** → ~42 days to collect payment.
- Indicates efficiency of **credit and collection policies**.

b. Inventory Turnover (INVT)

$$\text{INVT} = \frac{\text{Cost of goods sold}}{\text{Average inventory}}$$

- Measures how often inventory is sold and replaced.
- High INVT = strong inventory management, low holding costs.
- **PepsiCo's INVT (2020): 8.5 times** → Inventory turns roughly every **43 days**.
- Retailers (e.g., Walmart, Home Depot) usually have very high INVT ratios.

The DuPont Model

- Breaks down **Return on Equity (ROE)** into **three components** to explain *why* a company's ROE is high or low.

DuPont ROE decomposition

$$\text{ROE} = \underbrace{\frac{\text{Net Income}}{\text{Sales}}}_{\substack{\text{Profit Margin} \\ \text{(Profitability)}}} \times \underbrace{\frac{\text{Sales}}{\text{Average Total Assets}}}_{\substack{\text{Asset Turnover} \\ \text{(Efficiency)}}} \times \underbrace{\frac{\text{Average Total Assets}}{\text{Average Stockholders' Equity}}}_{\substack{\text{Financial Leverage} \\ \text{(Use of debt)}}}$$

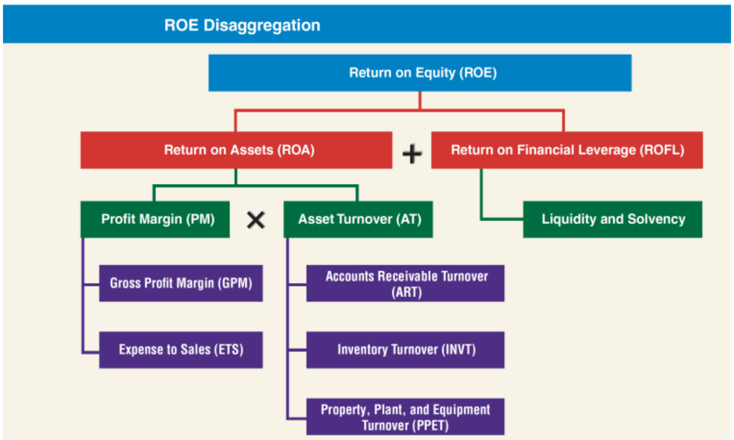
- ROE increases when a firm:
  - Earns **more profit** per sales dollar (↑ profit margin)
  - Uses **assets efficiently** to generate sales (↑ asset turnover)
  - Employs **more leverage** to amplify returns (↑ financial leverage)

- Useful for identifying whether performance comes from:
  - **Operational strength** (profitability, efficiency)
  - Or **financial structure** (use of debt)

Limitation

- Uses **net income**, not **earnings without interest expense (EWI)**.
  → So, **financial leverage affects both margin and ROE**, making it hard to isolate pure operating performance.[2]
- Despite this, it remains a **simple and widely used** framework for analyzing ROE.

ROE disaggregation identifies the two primary components of ROE – ROA and ROFL →



[2] **The traditional DuPont Model uses net income instead of earnings without interest expense (EWI)** in the profit margin component. Because net income includes interest expense, the model does not clearly separate operating performance from financing effects. As leverage increases, interest expense rises—reducing net income and profit margin—while at the same time increasing the financial leverage ratio and boosting ROE. This overlap means the model blends operating and financing effects, making it difficult to tell whether a higher ROE results from stronger operations or greater debt use. In short, the DuPont Model can misstate true operating efficiency when firms differ in their leverage levels.

Review Problems 1 – DuPont ROE Decomposition

Q4.8 Identifying the Unknown Companies. Presented below are common-sized income statement data for three well-known companies representing three different industries:

- Pharmaceutical
- Food and beverage
- Equipment manufacturing

Identify which of the three companies goes with each of the common-size data profiles. Discuss the rationale for your choice.

	Company 1	Company 2	Company 3
Net sales	100.0%	100.0%	100.0%
Cost of goods sold	28.8	29.7	45.4
Marketing, selling & admin.	20.0	38.8	37.1
Research & development	22.7	7.6	1.1
Other expenses	6.6	8.4	(3.0)
Net income	21.8%	15.5%	19.4%

Suggested Answer

Q4.10 Evaluating the Return on Equity. Presented below are selected financial data for two competitors. Discuss why the return on equity decreased for each firm.

	Return on Equity		Return on Sale		Asset Turnover		Financial Leverage	
	Year 1	Year 2	Year 1	Year 2	Year 1	Year 2	Year 1	Year 2
Company X	21.6%	16.7%	12.0%	8.0%	1.0	1.1	1.8	1.9
Company Y	24.9%	19.1%	9.9%	10.1%	1.2	0.9	2.1	2.1

Suggested Answer

Part 3: Liquidity and Solvency

10.6 How Debt Financing Can Increase a Firm's ROE

- Companies can use **debt (financial leverage)** to increase **Return on Equity (ROE)**.
- This effect is measured by **Return on Financial Leverage (ROFL)**
- ROE rises when the **return on assets (ROA)** exceeds the **cost of debt (interest rate)**.

Why Debt Boosts ROE?

- **Interest payments** to lenders are **tax-deductible**, reducing borrowing costs.
  - Example: At a 25% tax rate, a $1 interest costs the company only $0.75 after tax.
- **Debt is cheaper** than equity due to lenders' payment priority (i.e. less risky).
- When used wisely, debt lets a firm **generate more profit from shareholders' investment** without raising new equity.

Illustration: When Leverage Works for You – When the business is performing well.

	Company A (No Debt)	Company A' (No Debt)	Company A'' (50% Debt)
Assets (avg.)	**$500**	**$1,000**	**$1,000**
EWI (earnings before interest)	$50	$100	$100
Interest Expense (4%)	$0	$0	$20
Net Income	$50	$100	$80
Equity (avg.)	$500	$1,000	$500
ROA			
ROE			
ROFL			

Explanation

- Company A'' earns the same return on assets (10%) but uses half the equity.
- Its net income falls after interest, but because equity is smaller, its **ROE rises to 16%**.
- Thus, leverage magnifies shareholder returns.

When Leverage Works Against You – When the business underperforms.

- In other words, if the **cost of debt exceeds ROA**, leverage reduces ROE:

	Company A' (No Debt)	Company A'' (50% Debt)
Assets (avg.)	$1,000	$1,000
EWI (earnings before interest)	$30	$30
Interest Expense (4%)	$0	$20
Net Income	$30	$10
Equity (avg.)	$1,000	$500
ROA		
ROE		
ROFL		

Explanation

- Because interest (4%) exceeds ROA (3%), Company B's net income drops sharply.
- Leverage now **hurts ROE**, creating a **negative ROFL**.

Risk and Limitations of Debt Financing

- Higher debt increases **default risk** — the possibility that the firm cannot meet interest or principal payments.
- In downturns, leverage **amplifies losses**, making poor performance worse.
- Excessive debt can lead to **financial distress or bankruptcy**.

Debt Covenants

- Lenders often impose **covenants**—contractual restrictions to protect themselves.
  - Examples: limits on additional borrowing, dividend payments, or asset sales.
- These covenants act as an additional "cost" of debt financing.

Industry Differences

- Utilities: High debt tolerance (stable cash flows, regulated environment).
- Telecom, Chemicals: Moderate to high debt (large fixed-asset investment).
- Pharmaceuticals, Tech, Apparel: Lower debt (depend on intangible assets and R&D).

Liquidity vs. Solvency Analysis

- **Liquidity analysis:** Can the company generate enough **cash now** to pay its short-term obligations (interest, principal)?
- **Solvency analysis:** Can the company generate enough **cash in the long run** to stay out of bankruptcy?

10.7 Liquidity Analysis

- **Liquidity** measures how much cash a company has and how quickly it can raise cash to meet **short-term obligations**.
- Analysts use several ratios to evaluate liquidity:
  - Current Ratio (CR)
  - Quick Ratio (QR)
  - Operating Cash Flow to Current Liabilities (OCFCL)
  - Cash Burn Rate (CBR)

Current Ratio (CR)

$$\text{Current Ratio (CR)} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

- Indicates whether current assets can cover current liabilities.
- A ratio **above 1.0** suggests positive **working capital** (current assets > curr. liabilities).
- However, a ratio **below 1.0 isn't always bad**—some efficient firms operate successfully with lower ratios.

- **Retailers** (e.g., *Kroger*, *Walmart*) often have low current ratios since they turn inventory quickly and manage payables efficiently.
- **PepsiCo 2020:** CR = 0.98 (up from 0.86 in 2019) → indicates sufficient liquidity

Quick Ratio (QR)

$$\text{Quick Ratio (QR)} = \frac{\text{Cash + Short-term Securities + Accounts Receivable}}{\text{Current Liabilities}}$$

- Excludes inventory and prepaids; focuses on assets that can be **quickly converted to cash**.
- A **stricter measure** of short-term liquidity than the current ratio.
- **PepsiCo 2020:** QR = 0.77 (up from 0.66 in 2019).
  → Below 1.0 but typical for large manufacturing firms with predictable cash flows.

Operating Cash Flow to Current Liabilities (OCFCL)

$$\text{OCFCL} = \frac{\text{Cash Flow from Operations}}{\text{Average Current Liabilities}}$$

- Shows whether a company's **operations generate enough cash** to cover ST liabilities.
- A higher ratio means stronger ability to **pay debts using internal cash flow**.
- **PepsiCo 2020:** OCFCL = 0.48 (up slightly from 0.45 in 2019)
  → Reflects steady cash generation and improved efficiency, but long-term sustainability depends on future operations.

Cash Burn Rate (CBR)

$$\text{Cash Burn Rate (CBR)} = \frac{\text{Free Cash Flow in Period}}{\text{Number of Days}}$$

- Measures **how quickly a company uses its cash** when **free cash flow is negative**.
- Relevant for **startups** (e.g., *Tesla* in early years) or firms in **financial distress** (e.g., *Sears*).
- **Blue Apron 2020:**
  - Free cash flow = –$11.4M → burn rate = –$31K per day.
  - Indicates ongoing cash use, though improved from prior years.

Liquidity Analysis – Key Takeaways

- **Liquidity ratios** help assess whether a firm can **meet short-term obligations** without raising external funds.

- Firms must balance liquidity and profitability:
  - Too **low liquidity** → risk of default.
  - Too **high liquidity** → inefficient use of assets.

- A **combination of CR, QR, OCFCL, and CBR** provides a complete picture of short-term financial health.

10.8 Solvency Analysis

- **Solvency** measures a company's ability to meet **long-term debt obligations**, including both periodic **interest payments** and **repayment of principal**.
- A company that cannot meet these obligations becomes **insolvent** (fails financially).
- Solvency analysis helps assess a firm's **long-term financial stability** and **credit risk**.
- Two main approaches:
  1. **Balance sheet approach:** examines the proportion of debt versus equity financing.
  2. **Income statement approach:** examines how much profit is available to cover debt costs.

Debt-to-Equity Ratio (D/E)

$$\text{Debt-to-Equity Ratio} = \frac{\text{Total Liabilities}}{\text{Total Stockholders' Equity}}$$

- Measures how much the company relies on **debt financing** relative to **equity financing**.
- A **higher ratio** → greater reliance on debt, **lower solvency**, and **higher financial risk**.
- A **lower ratio** → more conservative capital structure and **greater financial stability**.

Example – PepsiCo 2020:
- D/E = 5.9 ($79,366M ÷ $13,552M)
- Up from 4.3 in 2019 — reflects more borrowing and share buybacks amid low interest rates.
- Industry average ≈ 1.31 → PepsiCo uses **substantially more debt** than most large firms.

Times Interest Earned (TIE)

$$\text{Times Interest Earned (TIE)} = \frac{\text{Earnings Before Interest and Taxes (EBIT)}}{\text{Interest Expense}}$$

- Indicates how many times the firm's **operating income** can cover its **interest payments**.
- A **higher TIE** means a company comfortably covers interest costs — **lower risk of default**.
- A **lower TIE** suggests potential difficulty meeting interest obligations.

Example – PepsiCo 2020:
- TIE = 8.24× = ($9,069M + $1,252M) ÷ $1,252M
- Down from 9.20× in 2019 but still strong, indicating **ample capacity to pay interest**.

Interpreting Solvency Ratios

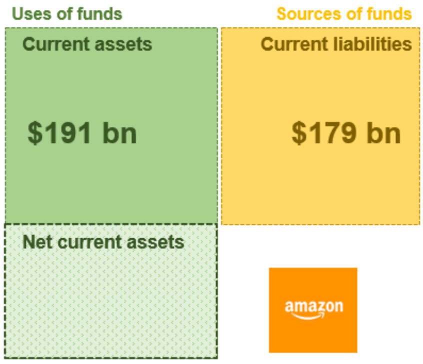
- **Debt-to-Equity (D/E):** Evaluates *capital structure* — how much funding comes from creditors versus shareholders.
- **Times Interest Earned (TIE):** Evaluates *interest coverage* — whether operating income can meet debt costs.
- Analysts use both to assess **financial leverage and long-term risk exposure**.
- A strong solvency position means:
  - The company can **sustain debt levels** and **meet obligations** over time.
  - There's **lower credit risk** and **greater borrowing capacity**.

10.9 Using Working Capital: A Source of Funds Perspective

This note reframes working capital not just as a measure of liquidity, but as a **requirement for long-term funding** that supports ongoing operations.

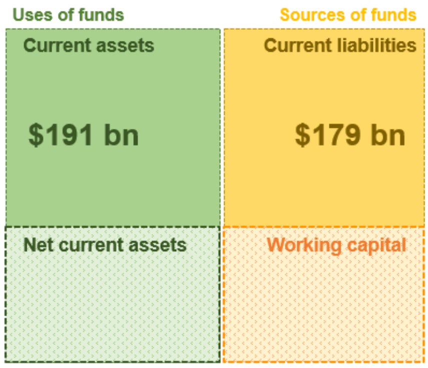
Traditional View – Working Capital and Liquidity

- Working capital management deals with efficient control of **cash and current assets** to maintain liquidity.
- Net Current Assets = Current Assets – Current Liabilities
- **Interpretation:**
  - A positive figure means the firm has enough current assets to pay current liabilities.
  - Indicates **short-term solvency** and operational stability.
  - Traditionally viewed as a **liquidity measure**, ensuring obligations can be met as they come due.



Source-of-Funds View – The Working Capital Requirement

- **Concept Shift -** Working capital can also be understood as the **amount of long-term financing required** to fund current operations.
- **Key Points**
  - Net current assets create a **working capital requirement [$12 bn (= $191-$179)]** .
  - It represents the **portion of current assets not financed by current liabilities**.
  - Hence, the business must use **long-term funds** (equity or long-term debt) to cover this shortfall.
  - In summary, when **current assets > current liabilities**, the gap must be financed by **long-term capital**—this is the true "working capital requirement."



Alternative Case – Net Current Liabilities

- Occurs when **current liabilities exceed current assets**.
- **Implications**
  - Short-term liabilities appear sufficient to fund current assets.
    - This can be seen as positive (to be elaborated a bit more later).
  - However, this position is often **financially unhealthy** and can signal **liquidity risk**.
  - The firm may struggle to meet short-term obligations.

- **Remedies**
  - Increase current assets by selling non-current assets, **or**
  - Raise **additional long-term capital** (equity or debt) to restore a positive working capital position.

The Working Capital Cycle – An Example :

Stage	Activity	Effect
Day 1 – Inventory Purchase	Buy $10K inventory on credit	↑ Inventory (CA) and ↑ Payables (CL)
Day 12 – Sale	Sell inventory (cost $6K) for $8K	Inventory ↓, Receivables ↑, profit ↑
Day 38 – Collection	Customer pays invoice	Receivables ↓, Cash ↑
Day 60 – Payment	Pay supplier	Cash ↓, Payables ↓

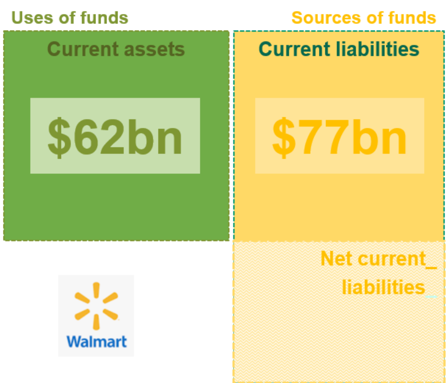
- This continuous loop requires maintaining a **stable amount of long-term funding** to cover the timing gap between payments and receipts.
- That stable amount = **the working capital requirement**.

Summary – Key Points

- **Positive working capital is healthy (but not always good):**
  - Having more current assets than current liabilities (a positive net current asset position) indicates good financial health and solvency.
  - It suggests the company can comfortably meet its short-term obligations.

- **Working capital is a funding requirement:**
  - Working capital represents the amount of **long-term financing** (through loans or equity) needed to bridge the gap between short-term assets and short-term liabilities.
  - However, **excessive positive working capital** isn't always ideal—it may signal that too much capital is tied up in low-return current assets.

- **Negative working capital is risk (but not always bad):**
  - When current liabilities exceed current assets, the company faces potential liquidity risk—it may struggle to pay its immediate bills.
  - Yet, in some business models, **negative working capital** can be efficient.
  - For example, companies like **Walmart** often operate with negative net current assets, which means they effectively use **vendors' capital** to finance their operations.



10.10 Limitations of Ratio Analysis

GAAP Limitations

- Measurability: Financial statements include only items that can be reliably measured. Intangible but valuable assets such as brand reputation or employee expertise are often excluded.
- Non-capitalized Costs: Many expenditures that add long-term value, like research and development (R&D) and advertising, are expensed rather than capitalized, leading to understated assets.
- Historical Costs: Assets are recorded at their original acquisition cost, not their current market value, which can distort ratios when values have changed significantly.

Company Changes

- Frequent mergers, acquisitions, divestitures, and strategic shifts make it difficult to compare ratios across periods or between companies.
- Life cycle stages and seasonal effects can also cause variations in ratios.

Conglomerate Effects

- Large corporations often operate across multiple industries or subsidiaries. Consolidated statements can obscure differences among segments, making it challenging to interpret ratios without segment-level analysis.

Means to an End

- Ratios condense complex operations into single numbers that cannot capture qualitative factors such as management effectiveness, innovation, or strategy.
- Successful analysis requires looking beyond ratios to understand a company's overall performance and direction.

10.11 Using Ratios to Predict Bankruptcy — Altman Z-Score (FYI only)

- Developed by **Professor Edward Altman (1968)** to predict the likelihood of **corporate bankruptcy**.
- Combines several financial ratios into a single **Z-score** to measure a company's **credit risk** and **financial health**.

Formula

$$Z = 1.2\left(\frac{\text{Working Capital}}{\text{Total Assets}}\right) + 1.4\left(\frac{\text{Retained Earnings}}{\text{Total Assets}}\right) + 3.3\left(\frac{\text{EBIT}}{\text{Total Assets}}\right) + 0.6\left(\frac{\text{Market Value of Equity}}{\text{Total Liabilities}}\right) + 0.99\left(\frac{\text{Sales}}{\text{Total Assets}}\right)$$

Interpretation of Variables

Variable	Measures	Meaning
Working Capital / Total Assets	Liquidity	Short-term financial flexibility
Retained Earnings / Total Assets	Profitability	Long-term reinvested earnings
EBIT / Total Assets	Efficiency	Operating profit relative to assets
Market Value of Equity / Total Liabilities	Leverage	Financial stability and capital strength
Sales / Total Assets	Turnover	Asset efficiency and revenue generation

Z-Score Interpretation

Z-Score Range	Financial Condition	Implication
> 3.0	Healthy company	Low bankruptcy risk
1.8 – 3.0	Gray zone	Moderate risk; needs monitoring
< 1.8	Distressed	High likelihood of bankruptcy

- Originally **95% accurate** one year before bankruptcy and **72%** two years ahead.
- Still widely used by **banks, investors, and credit analysts** for assessing **default risk**.
- Modern credit scoring systems often incorporate **Altman's Z-model** as a foundation.

10.12 Using Ratios Analyzing and Interpreting Core Operating Activities

Operating vs. Nonoperating Activities

- The analysis distinguishes between **operating** and **nonoperating activities** to better evaluate a company's profitability and efficiency. **Operating activities** are **central** to **value creation** and long-term sustainability.
- **Operating activities**: Core transactions needed to produce and deliver goods or services — such as sales, production, R&D, marketing, and customer service.
- **Nonoperating activities**: Financing and investing activities are important but not the company's primary value drivers.

Return on Net Operating Assets (RNOA)

- **RNOA** measures the return earned from **operating activities** relative to the **capital invested in those operations**.

$$RNOA = \frac{\text{Net Operating Profit After Taxes (NOPAT)}}{\text{Average Net Operating Assets (NOA)}}$$

- **NOPAT** focuses on **after-tax profit from operations only**, excluding financing and investing effects.

  NOPAT = NI – [(Nonoperating revenues – Nonoperating expenses) × (1 – Tax rate)]
- NOA represents the net capital tied up in operations and is calculated as operating assets minus operating liabilities.

Reporting Operating Activities in the Income Statement

- Operating accounts include: sales revenue, cost of goods sold, SG&A, R&D, depreciation, rent, insurance, and wages.
- Nonoperating accounts: interest income and expense, and gains/losses on investments
- **PepsiCo Example (2020)**

  NOPAT = $7,175 - [($117 - $1,128) × (1 - 0.25)] = $7,933.3

  NOPAT($7,933.3) is higher than net income ($7,175) due to interest expenses.

Reporting Operating Activities in the Balance Sheet

- Operating assets: Cash, receivables, inventories, PPE, and intangibles.
- Operating liabilities: Acct payable, accrued expenses, and employee benefit liability.
- Nonoperating items: Debt, investments in affiliates, and interest-bearing liabilities.
- **PepsiCo Example (2020)**

  NOA = $55,843 = Operating assets ($91,059) – Operating liabilities ($35,216).

Interpreting RNOA

- RNOA = NOPAT / Average NOA = $7,933.3 ÷ (($55,843 + $46,322) ÷ 2) = 15.5%.
- PepsiCo's RNOA of 15.5% indicates strong operating efficiency.
- Its ROE of 50.5% includes the effects of financial leverage, meaning 31% of total shareholder return comes from operating activities.

10.13Key Ratios - Formula

RETURN MEASURES

Return on Equity (ROE) $= \dfrac{\text{Net income}}{\text{Average stockholders' equity}}$

Earnings without Interest Expense (EWI) $= \text{N.I.} + [\text{Interest expense} \times (1 - \text{Statutory tax rate})]$

Return on Assets (ROA) $= \dfrac{\text{Earnings without interest expense (EWI)}}{\text{Average total assets}}$

Return on Financial Leverage (ROFL) $= \text{ROE} - \text{ROA}$

PROFITABILITY RATIOS

Profit Margin (PM) $= \dfrac{\text{Earnings without interest expense (EWI)}}{\text{Sales revenue}}$

Gross Profit Margin (GPM) $= \dfrac{\text{Sales revenue - Cost of goods sold}}{\text{Sales revenue}}$

Expense-to-Sales (ETS) $= \dfrac{\text{Individual expense items}}{\text{Sales revenue}}$

TURNOVER RATIOS

Asset Turnover (AT) $= \dfrac{\text{Sales revenue}}{\text{Average total assets}}$

Accounts Receivable Turnover (ART) $= \dfrac{\text{Sales revenue}}{\text{Average accounts receivable}}$

Inventory Turnover (INVT) $= \dfrac{\text{Cost of goods sold}}{\text{Average inventory}}$

Property, Plant, and Equipment Turnover (PPET) $= \dfrac{\text{Sales revenue}}{\text{Average total assets}}$

LIQUIDITY RATIOS

Current Ratio (CR) $= \dfrac{\text{Current assets}}{\text{Current liabilities}}$

Quick Ratio (QR) $= \dfrac{\text{Cash + Short-term securities + Accounts receivable}}{\text{Current liabilities}}$

Operating Cash Flow to Current Liabilities (OCFCL) $= \dfrac{\text{Operating cash flow}}{\text{Average current liabilities}}$

Cash Burn Rate $= \dfrac{\text{Free cash flow in the period}}{\text{Number of days in the period}}$

SOLVENCY RATIOS

Times Interest Earned (TIE) $= \dfrac{\text{Earnings before interest expense and taxes (EBIT)}}{\text{Interest expense}}$

Debt-to-Equity (D/E) $= \dfrac{\text{Total liabilities}}{\text{Total stockholders' equity}}$

Review Problems 1-5 Ratio Exercise – The Coca-Cola Company

THE COCA-COLA COMPANY AND SUBSIDIARIES
Consolidated Statements of Income
($ millions)

Year Ended December 31	2020	2019
Net operating revenues	$33,014	$37,266
Cost of goods sold	13,433	14,619
Gross profit	19,581	22,647
Selling, general, and administrative expenses	9,731	12,103
Other operating charges	853	458
Operating income	8,997	10,086
Interest income	370	563
Interest expense	1,437	946
Equity income (loss)—net	978	1,049
Other income (loss)—net	841	34
Income before taxes	9,749	10,786
Income taxes	1,981	1,801
Consolidated net income	$ 7,768	$ 8,985

THE COCA-COLA COMPANY AND SUBSIDIARIES
Consolidated Balance Sheets
($ millions)

December 31	2020	2019
Assets		
Current assets		
Cash and cash equivalents	$ 6,795	$ 6,480
Short-term investments	1,771	1,467
Marketable securities	2,348	3,228
Trade accounts receivable, net	3,144	3,971
Inventories	3,266	3,379
Prepaid expenses and other assets	1,916	1,886
Total current assets	19,240	20,411
Equity method investments	19,273	19,025
Other investments	812	854
Other assets	6,184	6,075
Deferred income tax assets	2,460	2,412
Property, plant, and equipment, net	10,777	10,838
Trademarks with indefinite lives	10,395	9,266
Goodwill	17,506	16,764
Other intangible assets	649	736
Total assets	$87,296	$86,381
Liabilities and equity		
Current liabilities		
Accounts payable and accrued expenses	$11,145	$11,312
Loans and notes payable	2,183	10,994
Current maturities of long-term debt	485	4,253
Accrued income taxes	788	414
Total current liabilities	14,601	26,973
Long-term debt	40,125	27,516
Other liabilities	9,453	8,510
Deferred income tax liabilities	1,833	2,284
Total liabilities	66,012	65,283
Total equity	21,284	21,098
Total liabilities and equity	$87,296	$86,381

THE COCA-COLA COMPANY AND SUBSIDIARIES
Consolidated Statements of Income
($ millions)

Year Ended December 31	2020	2019
Net operating revenues	100.0%	100.0%
Cost of goods sold	40.7%	39.2%
Gross profit	59.3%	60.8%
Selling, general, and administrative expenses	29.5%	32.5%
Other operating charges	2.6%	1.2%
Operating income	27.2%	27.1%
Interest income	1.1%	1.5%
Interest expense	4.4%	2.5%
Equity income (loss)—net	3.0%	2.8%
Other income (loss)—net	2.5%	0.1%
Income before taxes	29.4%	29.0%
Income taxes	6.0%	4.8%
Consolidated net income	23.4%	24.2%

THE COCA-COLA COMPANY AND SUBSIDIARIES
Common Size Balance Sheets

December 31	2020	2019
Assets		
Current assets		
Cash and cash equivalents	7.8%	7.5%
Short-term investments	2.0%	1.7%
Marketable securities	2.7%	3.7%
Trade accounts receivable, net	3.6%	4.6%
Inventories	3.7%	3.9%
Prepaid expenses and other current assets	2.2%	2.2%
Total current assets	22.0%	23.6%
Equity method investments	22.1%	22.0%
Other investments	0.9%	1.0%
Other assets	7.1%	7.0%
Deferred income tax assets	2.8%	2.8%
Property, plant, and equipment, net	12.3%	12.5%
Trademarks with indefinite lives	11.9%	10.7%
Goodwill	20.1%	19.4%
Other intangible assets	0.8%	1.0%
Total assets	100.0%	100.0%
Liabilities and equity		
Current liabilities		
Accounts payable and accrued expenses	12.8%	13.1%
Loans and notes payable	2.5%	12.7%
Current maturities of long-term debt	0.6%	4.9%
Accrued income taxes	0.9%	0.5%
Total current liabilities	16.8%	31.2%
Long-term debt	46.0%	31.9%
Other liabilities	10.8%	9.9%
Deferred income taxes	2.1%	2.6%
Total liabilities	75.7%	75.6%
Total equity	24.3%	24.4%
Total liabilities and equity	100.0%	100.0%

Review 5-1 Preparing Common-Size Income Statements and Balance Sheets
Above are summarized 2020 and 2019 income statements and balance sheets for The Coca-Cola Company, common-size income statements and balance sheets for Coca-Cola.
Required: Comment on any noteworthy relationships that you observe.

Review 5-2 Calculating ROE, ROA, and ROFL
Required: Refer to the financial statements for the Coca-Cola Company presented in Review 5-1. Calculate Coca-Cola's ROE, ROA, and ROFL for 2020. Assume a 25% statutory tax rate for this year.

Review 5-3 Disaggregating the Return on Equity Ratio into its Components
Required: Refer to the financial statements for the Coca-Cola Company

- Calculate Coca-Cola's profit margin (PM) and asset turnover (AT) ratios for 2020.
- Show that ROA = PM × AT using Coca-Cola's financial data.
- Calculate Coca-Cola's gross profit margin (GPM), accounts receivable turnover (ART), inventory turnover (INVT), and property, plant, and equipment turnover (PPET) ratios for 2020.
- Evaluate Coca-Cola's ratios in comparison to those of PepsiCo.

Review 5-4 Computing and Interpreting Liquidity and Solvency Ratios
Required: Refer to the income statements and balance sheets for the Coca-Cola Company. Compute the following liquidity and solvency ratios for Coca-Cola.

- Current ratio
- Quick ratio
- Debt-to-equity ratio
- Times interest earned

Review 5-5 Measuring the Effects of Operating Activities on ROE
Required: Refer to the financial statements of the Coca-Cola Company. For 2020, calculate Coca-Cola's return on net operating assets (RNOA) and then disaggregate RNOA into net operating profit margin (NOPM) and net operating asset turnover (NOAT). Assume a statutory tax rate of 25% and that equity method investments, other assets, and other liabilities are operating items.